DSA REFERENCE

Intervals Problem Ladder

A sequenced practice ladder from warm-up to interview stretch.

Sort ranges by a boundary, then reason about overlap using the last accepted or currently merged interval.

VERSION 1.0 REVIEW

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STATUS HUMAN REVIEW REQUIRED

DSA / INTERVALS

DSA

Intervals Problem Ladder

CORE IDEA

Sort ranges by a boundary, then reason about overlap using the last accepted or currently merged interval.

Recognition cues

+ merge or insert ranges
+ meeting conflicts
+ coverage
+ minimum rooms or removals

Implementation sequence

+ define inclusive versus half-open boundaries
+ choose sort key
+ compare next start with active end
+ merge or finalize
+ handle touching intervals deliberately

Invariant

Processed intervals are represented by a correct non-overlapping summary, and only the last summary interval can overlap the next sorted one.

Worked reasoning

When merging, if next.start <= current.end, extend current.end; otherwise emit current and start a new range.

Complexity

O(n log n) for sorting and O(n) for scanning; output may require O(n) space.

Failure modes

- [ ] wrong overlap inequality
- [ ] sorting by the wrong endpoint
- [ ] mutating caller data unexpectedly
- [ ] confusing room-count and merge logic

Practice ladder

Intervals Problem Ladder: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Merge Intervals**	State the cue, invariant, edge case, and proof before implementation.
2	**Insert Interval**	State the cue, invariant, edge case, and proof before implementation.
3	**Non-overlapping Intervals**	State the cue, invariant, edge case, and proof before implementation.
4	**Meeting Rooms**	State the cue, invariant, edge case, and proof before implementation.
5	**Meeting Rooms II**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.